Benedetta Mariani

Computational neuroscience · Physics of complex systems

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Padua, Italy | Italian | Born 21 December 1995

Professional Profile

Computational neuroscientist with a PhD in Neuroscience and a background in the physics of complex systems. Research experience spans dynamical systems, neural data analysis (neural oscillations, neural avalanches, effective connectivity), machine learning and multimodal neural data analysis. Experienced in communicating interdisciplinary research through publications, talks and teaching.

Work Experience

Oct 2024 – Oct 2026

Postdoctoral Researcher

Padova Neuroscience Center · University of Padua, Italy
Inference of directed brain interactions from fMRI data using dynamic causal modeling and state-space approaches. Characterization of latent brain dynamics and patient stratification with Hidden Markov Models. Development of predictive models of clinical and behavioral outcomes.
Supervisors: Prof. Michele Allegra and Prof. Samir Suweis.

Oct 2020 – Sep 2024

PhD Researcher in Neuroscience

Padova Neuroscience Center / Laboratory of Interdisciplinary Physics · University of Padua, Italy
Combined complex-systems physics, computational modelling and neural data analysis to investigate collective behaviour in neural and biological systems.
Studied how dynamical regimes and oscillatory activity shape computation in neural systems.
Supervisor: Prof. Samir Suweis.

Dec 2019 – Sep 2020

Research Fellow

Laboratory of Interdisciplinary Physics · University of Padua, Italy
Project: "Criticality in rat somatosensory barrel cortex".
Supervisors: Prof. Samir Suweis and Prof. Stefano Vassanelli.

Education And Training

2020 – 2025

PhD in Neuroscience

University of Padua · Padua, Italy
Research conducted at the Padova Neuroscience Center and Laboratory of Interdisciplinary Physics.
Degree awarded in January 2025.

Dec 2017 – Oct 2019

MSc in Physics of Complex Systems

University of Turin · Turin, Italy
Final grade: 110/110 cum laude.
Thesis: "Signatures of criticality in rat brain: beyond avalanche statistics".
Research performed at the Laboratory of Interdisciplinary Physics and NeuroChip Laboratory, University of Padua.

Sep 2015 – Dec 2017

BSc in Physics

University of Padua · Padua, Italy
Final grade: 100/110.
Thesis: "Phase separation for intrinsically disordered proteins".

Oct 2014 – Sep 2015

Studies in Biomedical Engineering

Polytechnic University of Milan · Milan, Italy

Research Interests And Expertise

Research areas: Computational neuroscience; dynamical systems; complex systems; neural oscillations and criticality; neural coding; machine learning for neural data; learning representations in neural data.

Data and methods: electrophysiology, neuronal spike trains, resting-state fMRI, functional and effective connectivity, dynamic causal modeling, state-space modeling, Hidden Markov Models, stochastic and whole-brain modelling, graph and network analysis, statistical inference and machine learning.

Technical tools: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch, Matplotlib, nilearn/nibabel), MATLAB/SPM, R, C++, Mathematica, Linux/Unix, high-performance computing and Slurm.

Peer-Reviewed Publications

2026 **Whisker stimulation reinforces a resting-state network in the barrel cortex: Nested oscillations and avalanches**
B. Mariani, R. Guevara, M. Tambaro, M. Maschietto, A. Leparulo, S. Vassanelli, S. Suweis
PLOS Computational Biology, 22(7), e1014521. DOI: 10.1371/journal.pcbi.1014521

2023 **Prenatal experience with language shapes the brain**
B. Mariani, G. Nicoletti, G. Barzon, M. C. Ortiz Barajas, M. Shukla, R. Guevara, S. S. Suweis, J. Gervain
Science Advances, 9(47), eadj3524. DOI: 10.1126/sciadv.adj3524

2022 **Criticality and network structure drive emergent oscillations in a stochastic whole-brain model**
G. Barzon, G. Nicoletti, **B. Mariani**, M. Formentin, S. Suweis
Journal of Physics: Complexity, 3, 025010. DOI: 10.1088/2632-072X/ac7a83

2022 **Disentangling the critical signatures of neural activity**
B. Mariani, G. Nicoletti, M. Bisio, M. Maschietto, S. Vassanelli, S. Suweis
Scientific Reports, 12, 10770. DOI: 10.1038/s41598-022-13686-0

2021 **Neuronal avalanches across the rat somatosensory barrel cortex and the effect of single whisker stimulation**
B. Mariani, G. Nicoletti, M. Bisio, M. Maschietto, R. Oboe, A. Leparulo, S. Suweis, S. Vassanelli
Frontiers in Systems Neuroscience, 15, 709677. DOI: 10.3389/fnsys.2021.709677

Preprints

2026 **Towards a scalable effective connectivity estimation in fMRI**
B. Mariani, L. Taffarello, G. Baron, G. Barzon, F. Calesella, M. Gilson, S. De Marchi, A. Chiuso, A. Vallesi, S. Suweis, M. Zorzi, A. Bertoldo, M. Allegra.
bioRxiv, preprint. DOI: 10.64898/2026.10.26.761780 | bioRxiv
Software: mOU-EC pipeline | Source code

2026 **Discovery of Latent Clinical Phenotypes Through Optimal Transport of Brain Dynamics**
L. Taffarello, F. Calesella, G. Barzon, **B. Mariani**, A. Gallo, M. Altieri, A. P. De Rosa, A. D'Ambrosio, F. Esposito, S. De Marchi, A. Vallesi, M. Allegra, A. Bertoldo, S. Suweis, M. Zorzi.
Preprint manuscript. Full manuscript

2026 **Brain-tuning near criticality in newborns by prenatal experience with language**
Jesus Encinas, **Benedetta Mariani**, Ramon Guevara, Maria Ortiz-Barajas, Judit Gervain, Samir Suweis, Fabrizio Lombardi.
bioRxiv, preprint. DOI: 10.64898/2026.05.21.726940 | bioRxiv

2025 **The somatosensory barrel cortex controls the spindle thalamocortical oscillation by frequency locking**
M. Tambaro, R. Guevara, M. Maschietto, A. Leparulo, C. Cecchetto, G. Nicoletti, **B. Mariani**, S. Suweis, S. Vassanelli.
bioRxiv, preprint. DOI: 10.1101/2025.07.09.662963 | bioRxiv

Teaching And Supervision

Oct 2025

Teaching Assistant – Advanced Machine Learning

MSc in Quantitative and Computational Biosciences · University of Padua
Topics: deep feedforward neural networks, convolutional and recurrent neural networks, graph convolutional networks and generative models.
Module coordinator: Prof. Nicolò Navarin.

Dec 2022

Co-supervisor of BSc thesis in Physics

University of Padua · Padua, Italy
Thesis: "Analysis of neural oscillations and synchrony in the brain".

Selected Talks And Conference Contributions

Jun 2026

OHBM 2026, Bordeaux, France

Towards consistent and scalable whole-brain effective connectivity estimation.

Mar 2026

COSYNE 2026, Lisbon, Portugal

Self-sustained oscillations link spontaneous and stimulus-driven dynamics in the barrel cortex.

Oct 2025

Bernstein Conference 2025, Frankfurt, Germany

Nested oscillations and critical avalanches prime the rat barrel cortex for whisker stimuli.

Sep 2025

INCTN 2025, Padua, Italy

Poster: Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches.

Jul 2025

Annual Computational Neuroscience Meeting (OCNS), Florence, Italy

Spontaneous oscillations and neural avalanches are linked to whisker-stimulation responses in the rat barrel–thalamus circuit.

Jun 2025

StatPhys29 Satellite, Venice, Italy

Contributed talk: Whisker stimulation reinforces a resting-state network in the barrel cortex: nested oscillations and avalanches.

Feb 2025

SIFS Young Seminar Series

Invited talk: Spontaneous oscillations and avalanches are linked to whisker sensory-response processing in the rat thalamocortical circuit.

May 2024

Baby Rhythm ERC Closing Workshop, Padua, Italy

Invited talk: Prenatal experience with language shapes the brain.

Sep 2022

Bernstein Conference 2022, Berlin, Germany

Collective oscillations in the rat barrel–thalamus network; co-author of a second contribution on critical signatures of neural activity.

Sep 2021

Bernstein Conference 2021, online

Disentangling the role of external and intrinsic dynamics in the critical signatures of neural activity.

Oct 2020

Brain Criticality Virtual Meeting

Contributed spotlight talk: Brain criticality beyond resting-state neuronal avalanches.

Oct 2020

Bernstein Conference 2020, online

Brain criticality beyond resting-state neuronal avalanches.

Conference Organisation

Sep 2025

Local Organising Committee member

Italian Network for Computational and Theoretical Neuroscience (INCTN) 2025 · Padua, Italy

Selected Schools And Additional Training

May 2024 — **Stochastic Models and Experiments in Ecology and Biology, Gran Sasso Science Institute, L'Aquila, Italy**

Mar 2022 — **COSYNE tutorials and main meeting, Lisbon, Portugal**

Sep 2021 — **EITN Fall School in Computational Neuroscience, Paris, France — project group on mean-field and population models; training in Brian, NEST, NEURON and The Virtual Brain.**

May–Jun 2021 — **Beg Rohu Summer School, Saint-Pierre-Quiberon, France**

Feb 2021 — **Computational and Systems Neuroscience (COSYNE), online**

Dec 2020 — **BrainHack Padova, online — led a project on white-matter microstructure and personality traits using Human Connectome Project data.**

Sep 2019 — **Summer School in Computational and Theoretical Models in Neuroscience, European Centre for Living Technology, Venice, Italy**

Awards, Distinctions And Public Engagement

2023 — **International media coverage for "Prenatal experience with language shapes the brain"**
The study received coverage from international scientific and general-interest media, including Le Scienze, Nature Italy, El País, The Times, Le Figaro and New Scientist, as well as science podcasts.

2022 — **Scientific Reports Top 100 Neuroscience papers of 2022**
"Disentangling the critical signatures of neural activity" was included among the journal's Top 100 Neuroscience papers of the year.

2021 — **SECS Grant**
Scholarship for Events on Complex Systems awarded by the Young Researchers of the Complex Systems Society.

2014 — **Italian Ministry of Education merit award**
Award for secondary-school graduates achieving the highest final grade with honours.

Language Skills

Italian Native — Mother tongue
English Fluent — Cambridge First Certificate in English
German Basic level — Goethe-Institut, Bremen